Ghassaei's Leg mechanism

Ghassaei's Linkage

Designed by Amanda Ghassaei, this is an optimized crank-based 8-bar leg mechanism. It is an improvement over Jansen's linkage in terms of design simplicity, symmetry of locus/foot trajectory, straightness of the trajectory in the stride phase, etc.

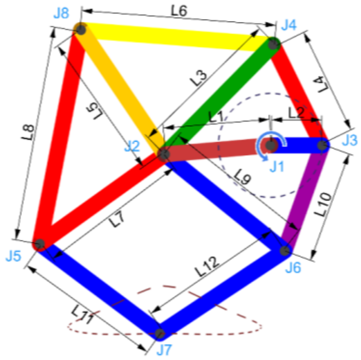

Figure 1: *Ghassaei's linkage in MechAnalyzer showing its coupler curve*

Walking Mechanisms

- Biped robots present certain challenges due to the large number of actuators: complexities of design & control, low energy efficiency, overly high price complicating practical use.
- A different way to think about biped robots is to use reduced degrees of freedom by way of using a leg mechanism, which can simulate the walking gait of humans an animals.
- Chebyshev was one of the first to suggest a walking machine based on the fourbar linkage: *the Pantigrade Machine*. (Original models available at the Museum of the History of Physics and Mathematics of Saint Petersburg State University, Peterhof, Russia).
- Chebyshev's machine spawned the development of many different leg mechanisms, using many different standard mechanisms as their bases: *the fourbar linkage with a pantograph, the Watt-II linkage, the Stephenson-III linkage, the Klann linkage, the Jansen linkage etc.*

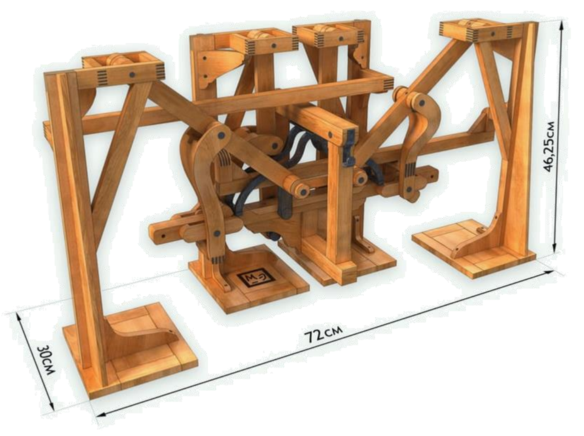

Figure 2: *Chebyshev's Pantigrade Walker*